

Enhanced Supervised Descent Learning Technique for Electromagnetic Inverse Scattering Problems by the Deep Convolutional Neural Networks

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Abstract—This work proposes a novel deep learning (DL) framework to solve the electromagnetic inverse scattering (EMIS) problems. The proposed framework integrates the complex-valued deep convolutional neural network (DConvNet) into the supervised descent method (SDM) to realize both off-line training and on-line “imaging” prediction for EMIS. The offline training consists of two parts: 1) DConvNet training: the training dataset is created, and the proposed DConvNet is trained to realize the EM forward process and 2) SDM training: the trained DConvNet is integrated into the SDM framework, and the average descent directions between the initial prediction and the true label of SDM iterative schemes are learned based on the same dataset in part 1). In the online step, the contrasts (permittivities) reconstruction of scatterers is realized by the SDM iteration process based on learned descent directions, while its forward process is achieved by the trained complex-valued DConvNet. Ultimately, this framework provides a new perspective to integrate the prior information into the EMIS solving process with the maintained accuracy. Unlike the conventional SDM, the novel proposed framework can significantly shorten the computation and realize the real-time imaging. Various numerical examples and discussions are provided to demonstrate the efficiency and accuracy of the proposed novel framework.

Index Terms—Convolutional neural network, deep learning (DL), electromagnetic inverse scattering (EMIS), supervised descent method (SDM).

I. INTRODUCTION

THE electromagnetic inverse scattering (EMIS) analysis is essential for microwave imaging [1], [2]. Its objective is to investigate parameters from the measured scattered field information and obtain images of scatterers’s certain properties [1], [2], [3]. Because of its noncontact and nondestructive detection features, EMIS has been extensively used in subsurface sensing [4], microwave remote sensing [5], biomedical imaging [6], etc.

During the past few decades, a number of approaches have been proposed to solve the EMIS problem, such as Born iterative method [7], contrast source inversion [8], [9], contrast source extended Born [10], and subspace optimization method [11]. Unfortunately, nearly all the conventional methods suffer limitations by the unavoidable nonlinearity and ill-posedness [1], [2], [3], [12]. In fact, to stabilize the EMIS solving process and decrease the nonuniqueness, the conventional methods usually use regularization techniques that contain prior information [12]. Unfortunately, the incorporated prior information is quite limited, because some information is difficult to be expressed in rigorous mathematical formulations [12]. To optimize the objective function, the gradient-based deterministic methods are usually used in the conventional approaches for EMIS problems, including Gauss–Newton methods and conjugate gradient methods [13]. These gradient-based deterministic methods iteratively minimize their corresponding objective functions by computing partial derivatives with respect to the model parameters in each optimization iteration, which unavoidably leads to longer time and higher memory cost. Moreover, the derivatives of the objective function can only illustrate the local property rather than the global one, which makes the optimization possibly trapped around the local minima. On the other hand, global optimization techniques such as genetic algorithm [14], differential evolution algorithm [15], or particle swarm optimizer [16] can be the alternatives for solving inverse scattering problems. But these methods are even more computational expensive than the deterministic approaches, making it hard to be used for real-time imaging.

Manuscript received 31 March 2020; revised 21 December 2020; accepted 26 January 2021. Date of publication 16 August 2022; date of current version 8 September 2022. This work was supported in part by the Research Grants Council of Hong Kong under Grant GRF 17207114, Grant GRF 17210815, Grant GRF 12200317, Grant GRF 12300218, Grant GRF 12300519, and Grant GRF 17201020; in part by the Asian Office of Aerospace Research and Development (AOARD) under Grant FA2386-17-1-0010; in part by the Hong Kong UGC under Grant AoE/P-04/08; in part by the National Science Foundation of China (NSFC) under Grant 61271158, Grant 61571264, and Grant 61971623; in part by the National Key Research and Development Program of China under Grant 2018YFC0603604; in part by the Guangzhou Science and Technology Plan under Grant 201804010266; in part by the Beijing Innovation Center for Future Chip; and in part by the Research Institute of Tsinghua, Pearl River Delta. (Corresponding authors: Maokun Li; Michael Kwok Po Ng.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TAP.2022.3196496>.

Digital Object Identifier 10.1109/TAP.2022.3196496

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Machine learning (ML) is rapidly progressing in modern computational science [17], [18]. There have been a large number of innovated applications, such as scientific computation [19], [20], [21], remote sensing imaging [22], [23], and hybrid field-circuit simulation [24], [25]. Artificial neural network (ANN) has been used to solve EMIS problems in [26] and [27]. Moreover, learning-by-examples (LBE) techniques has also been applied to various EM problems [28]. In addition, the support vector machine (SVM) has also been applied for nondestructive evaluation [29], [30]. Recently, a number of works based on deep learning (DL) have been reported [31], [32], [33], [34], [35]. At present, a number of DL methods are correlated with the conventional methods and need to use the initial inputs provided by traditional methods for further operation, such as initial inputs from the backpropagation (BP) method [36]. Also, most of the DL methods build the nonlinear relationship between the measured data and model parameters as a “blackbox” process, such as [32], [33], [34], and [35], which is potentially enhanced by the physical information provided by forward modeling. Meanwhile, the computational cost of DL approaches still has potential to be improved for online imaging.

In [37], supervised descent method (SDM) has been proposed to solve the EMIS problem with satisfactory accuracy and generalization ability. The SDM overcomes the difficulties of the conventional methods and incorporates more prior information into the solving process to accelerate and improve inversion. However, it has to compute the EM forward process in each optimization iteration in both the off-line and online computation processes. Thus, it has to construct EM coupled equations and calculate the complex Green’s functions [38], [39]. Hence, the high computational complexity and time cost cannot fulfill the requirement of real-time microwave imaging.

In this work, we propose a novel ML framework for EMIS by integrating the complex-valued deep convolutional neural network (DConvNet) into SDM. The application of the proposed approach consists of two steps: the off-line training step and the online prediction step. In the offline training step, the proposed DConvNet is trained to model the EM forward process, while the average descent directions are learned from the data offered by DConvNet to realize SDM for the optimization iteration process. In the online step, the permittivities (contrasts) of targets can be updated using both the prelearned DConvNet for the EM forward process and the prelearned descent directions for the optimization iteration process.

The advantages of the proposed ML framework are: 1) more prior information can be integrated in a flexible manner to improve the performance of the proposed method, because the training data in the off-line step can be created according to prior information; 2) compared with the conventional SDM [37], the computation cost for creating training data in the proposed ML framework is not increased, because the training of DConvNet and the training of SDM share the same set of training data; 3) after training, the proposed method does not need to compute the partial derivatives. Hence, the optimization iteration process could be greatly accelerated; 4) the computation complexity of the proposed method has been greatly reduced, because using DConvNet greatly

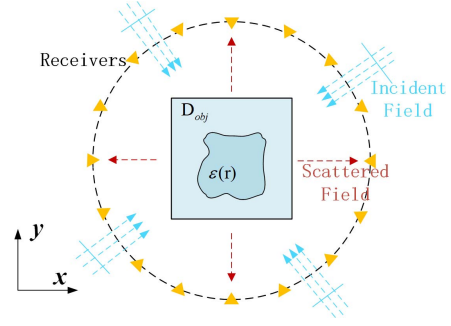


Fig. 1. TM_z wave scattering for the region D_{obj} .

improves the efficiency of the forward solving process. Thus, it can realize microwave imaging in real-time; and 5) thanks to sufficient prior information, the accuracy of the proposed method is better than the conventional approaches in solving the EMIS problems.

This article is organized as follows. In Section II, the formulation of EMIS is demonstrated, while the newly proposed DConvNet is described. The integration of DConvNet into SDM is also presented in Section II. In Section III, various numerical benchmarks are offered to demonstrate the validity and accuracy of the proposed method. Finally, the conclusion is given in Section IV.

II. THEORY AND FORMULATION

A. EM Forward and Inverse Modeling

Fig. 1 demonstrates the working mechanism for the EMIS process. There are M receivers equally spaced around the object domain D_{obj} . The TM_z incident wave E^{in} illuminates the object dielectric domain D_{obj} , which is discretized as $N \times N$ uniform fragments. The M receivers receive the scattered EM fields E^s . Although Fig. 1 is the typical 2D scenario to demonstrate the mechanism of the proposed approach, it should be highlighted that this approach can work for 3D scenarios. Two governing equations, i.e., Lippmann–Schwinger equations [40], are introduced to formulate the process of EMIS.

The interaction between the fields in fragments of D_{obj} can be formulated by the first equation, as described in (1):

$$E^t(\mathbf{r}) = E^{in}(\mathbf{r}) + k_0^2 \int_{D_{obj}} G(\mathbf{r}, \mathbf{r}') \chi(\mathbf{r}') E^t(\mathbf{r}') d\mathbf{r}' \quad (1)$$

where $G(\mathbf{r}, \mathbf{r}')$ represents the Green’s function and $G(\mathbf{r}, \mathbf{r}') = -(j/4)H_0^{(2)}(k_0|\mathbf{r}-\mathbf{r}'|)$ for the 2-D TM_z wave. $H_0^{(2)}$ is denoted as the second kind zeroth order Hankel function, and k_0 stands for the wave number in the free space. $\mathbf{r}' = (x', y')$ and $\mathbf{r} = (x, y)$ are the source points and field points in D_{obj} , respectively. The total EM field in D_{obj} is denoted E^t , while the contrast function is represented as $\chi(\mathbf{r}') = \varepsilon_r(\mathbf{r}') - 1$.

To describe the EM scattered field E^s created by the total field E^t in D_{obj} , the second equation is used to describe the interaction between E^s and E^t :

$$E^s(\mathbf{r}) = k_0^2 \int_{D_{obj}} G(\mathbf{r}, \mathbf{r}') \chi(\mathbf{r}') E^t(\mathbf{r}') d\mathbf{r}' \quad (2)$$

where $\mathbf{r} = (x_R, y_R)$ and $\mathbf{r}' = (x', y')$ are the positions of the receivers and the positions of source fragments, respectively.

The aim of solving EMIS problem is reconstructing the contrast χ (or permittivity ϵ_r) of target scatterers. The received EM scattered field provided by M receivers surrounding D_{obj} is utilized to realize this aim. However, reconstructing the contrasts by directly solving the mentioned two equations is very difficult [1], [3]. Usually, as presented in (3), the traditional methods create the objective function $f(\chi)$, based on which the traditional methods reconstruct the contrast by optimization.

$$\min : f(\chi) = \sum_{i=1}^{N_i} \|E_i^s - E_i^s(\chi)'\|^2 + \alpha T(\chi) \quad (3)$$

where N_i incident EM field E^{in} , respectively, produce the scattered field E_i^s received by the receivers. Based on (3), $E_i^s(\chi)'$ (i.e., the optimized EM scattered field) is gradually optimized to E_i^s by iterative computation. While $\|\cdot\|$ represents the Euclidian length, while T is the regularization to balance data fitting and solution stability [34], [41].

The specific solution process for most of conventional methods can be summarized as: First, (1) and (2) are discretized into (4) and (5) [1], [3], where D_{obj} is uniformly divided into pieces with their contrast and total EM field as piece-wise constant. Second, the coupled equations (4) and (5) are solved by iterative optimization computation in (3).

$$\bar{E}^s = \bar{\bar{G}}_R \cdot \text{diag}(\bar{E}^t) \cdot \bar{\chi} \quad (4)$$

$$\bar{E}^t = \bar{E}^{in} + \bar{\bar{G}}_D \cdot \text{diag}(\bar{E}^t) \cdot \bar{\chi} \quad (5)$$

where the matrices $\bar{\bar{G}}_R$ and $\bar{\bar{G}}_D$ have size $M \times N^2$ and $N^2 \times N^2$, respectively. We have $\bar{\bar{G}}_R = k_0^2 S_{n'} G(\mathbf{r}_n, \mathbf{r}_{n'})$ and $\bar{\bar{G}}_D = k_0^2 S_{n'} G(\mathbf{r}_n, \mathbf{r}_{n'})$, where $S_{n'}$ is the area of the source piece. We also have $n_r = 1, \dots, M$, $n = 1, \dots, N^2$, and $n' = 1, \dots, N^2$.

Unluckily, these traditional methods based on optimization also encounter challenges to solve (3), due to nonlinearity [1], [3]. Also, they can hardly work for the high-contrast cases [1], [3].

B. DConvNets for EM Forward Process

In this section, a novel DL approach is proposed based on the complex-valued DConvNet to efficiently model the EM forward process. In this novel model, DConvNet is used to transform the original contrasts (permittivities) of scatterers in the objective region into the corresponding scattered fields received by the receivers in Fig. 1 in Section II-A.

The proposed DConvNet is revised based on the structure of super-resolution convolutional neural network (SRCNN) [42], which has been widely used for image regression. Based on the structure of SRCNN [42], the proposed DConvNet has a number of appealing properties. To begin with, based on SRCNN, the structure of the proposed DConvNet is intentionally designed with simplicity but demonstrates superior accuracy [42]. Besides, based on SRCNN with moderate numbers of filters and layers, this structure can achieve fast speed for practical online usage [42].

The specific internal structure of our DConvNet is presented in Fig. 2. Its output is the scattered field E^s , referred to as

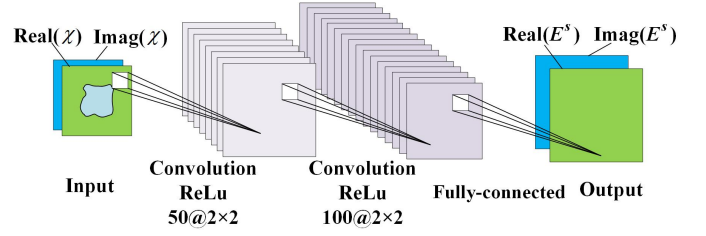


Fig. 2. Proposed DConvNet architecture for the EM forward process.

TABLE I
DCONVNET ARCHITECTURE

Type	Filter Number	Filter Size	Stride	Input Size	Output Size
Convolution	50	2×2	[2, 2]	$N \times N \times 2$	$(N/2) \times (N/2) \times 50$
ReLU				$(N/2) \times (N/2) \times 50$	$(N/2) \times (N/2) \times 50$
Convolution	100	2×2	[1, 1]	$(N/2) \times (N/2) \times 50$	$(N/2) \times (N/2) \times 100$
ReLU				$(N/2) \times (N/2) \times 100$	$(N/2) \times (N/2) \times 100$
Fully-connected				$(N/2) \times (N/2) \times 100$	$N \times N \times 25$
Regression Loss function				$N \times N \times 25$	$M \times N_i \times 2$

“field data,” which is the matrix with the size of $M \times N_i \times 2$. M represents M receivers and N_i is the total number of the incident field, while the real and imaginary parts of E^s stand for its two tubes, which is presented in Fig. 2. Meanwhile, its input is an $N \times N \times 2$ matrix comprising the real and imaginary parts of the target permittivity.

The proposed DConvNet is equipped with repetitive convolution and rectified linear unit (ReLU) [42], [43], [44] and the final fully connected layer. In the proposed DConvNet, the convolutional layer and the activation layer unit operate to capture features from input scatterer contrast (permittivity) χ and reorganize the features layer by layer until the final E^s is generated. The specific convolutional layer number, kernel number, its size, and the stride for kernel of the structure for this step are shown in Table I. Then, the convolutional layer and the activation layer unit feed into a final fully connected layer, which predicts the scattered field E^s . Through the proposed DConvNet, the original χ is transformed into the final “field data” E^s as a “heterogeneous” transformation process. This final output is used to calculate the mean squared error (mse) between the predicted value and the true label, defined as the loss

$$\text{loss} = (1/P) \sum_{i=1}^P (E_{\text{target},i}^s - E_{\text{predict},i}^s)^2 \quad (6)$$

where P is the number of responses. i indexes into each pixel of the target $E_{\text{target},i}^s$ and prediction $E_{\text{predict},i}^s$.

The nonlinear relationship between the input and the output is denoted as (7), while F is denoted as the nonlinear operation of the proposed DL model

$$\begin{aligned} E^s &= F(\chi) \\ &= w_{N_c} \sigma_{N_c-1}(\dots(w_2 * \sigma_1(w_1 * \chi + d_1) + d_2) \dots) + d_{N_c} \end{aligned} \quad (7)$$

where w_k are the parameters of the convolution kernel (filter weights) for the convolutional layers or the weight matrix

for the fully connected layers of the network at level k ($k = 1, \dots, N_C$, and $N_C = 3$ for the model in Fig. 2), while d_k are the bias terms. The function σ_k comprises the ReLU [32], [33], [34], [35], [42], [43], [44], [48]. The precise model architecture is shown in Table I.

The entire process is summarized as a “heterogeneous” process. It transforms contrast of the target scatterers into the final received scattered field. In the process, the complex-valued DConvNet directly uses χ as the input, while the output is chosen as E^s . Besides, we highlight that the whole process of the Green’s function and reduces the computation cost of the conventional methods [7], [8], [9], [10], [11]. Moreover, our DConvNet is used to realize the EM forward process. Thus, it will not encounter the ill-posedness of the EM inverse process [31], [32], [33], [34], [35], because the EM field generated by the certain incident wave can be uniquely identified by Maxwell’s equations in the EM forward process [1], [48]. In addition, in the online application process, the conventional methods to compute E^s from χ in (7) generally suffer from high computation complexity, because the conventional methods, such as Gauss–Newton method, has to compute the EM forward process during the optimization process, which means the computation costing process for solving matrix inversion, as shown in (4) and (5). However, the proposed DConvNet model successfully avoids computing the matrix inversion and greatly improves the efficiency. Thus, it is suitable for real-time application. In addition, unlike [34] and [35], the proposed DConvNet directly realizes the EM forward process, i.e., the “heterogeneous” transformation from contrast to scattered field and does not need the skip connection in the U-Net to realize residual learning.

There are several issues about the proposed model above that need to be emphasized.

- 1) *Filtering in Multichannel*: Filters exerted on multichannels act as the standard approach for ConvNet to augment the power of the structure [32], [33], [34], [35], [42], [43], [44], [48], based on which there are a number of feature maps at each layer. In fact, multichannel filtering in our DConvNet can be regarded as an analogous operation in the conventional iterative methods, such as partitioning wavelet coefficient to wavelet subband for ISTA formulation [41], [49].
- 2) *Complex-Valued Input and Output*: As a representative complex-valued problem, both the received electronic field and the contrast of the object domain are complex-valued for the EMIS problem. Thus, the original channel number of both the input and the output of the SRCNN has to be revised [42], so that their real and imaginary parts could be adapted in different channels. Hence, the novel model can be more adaptable and flexible for the EM forward process for real application scenario.
- 3) *Computational Complexity*: The DConvNet has the input with the size of $N \times N \times 2$, R filters with the size of $K_i \times K_i$ in each layer and f layers. Moreover, the final fully connected layer maps the last convolution and ReLU to the output contrast of the target scatterer with the size of $M \times N_i \times 2$. Generally, while the computational complexity of DConvNet is mainly decided by

convolutional operation [48], [49], we tend to choose the size of input contrast “image” to be approximated to the size of the output field “image,” as shown in [34], [35], and [48]. Hence, the computational complexity of the entire network can be approximated to $O(N^2 K_i^2 R^2 f)$ [48], [49]. Besides, its storage mainly depends on the size of the filters and biases. Thus, the storage of the proposed DConvNet is $O(K_i^2 R^2 f)$ [48], [49].

C. SDM for EM Inverse Process

In this section, the solving process of the EMIS problem by the SDM framework is explained in detail [37], [50]. The inversion procedure can be described as a typical optimization process [37], [50], which optimizes the unknown parameters of the model by minimizing the data misfit between the observed data E_{obs}^s and computed data E_{com}^s , as shown in the following equation:

$$S(\chi) = \|E_{com}^s - E_{obs}^s\|_2^2. \quad (8)$$

Equation (8) can be expanded by the Taylor expansion in the second order in $\chi = \chi_0 + \Delta\chi$ as follows:

$$S(\chi_0 + \Delta\chi) \approx S(\chi_0) + \mathbf{J}_S(\chi_0)^T \Delta\chi + \frac{1}{2} \Delta\chi^T \mathbf{H}_S(\chi_0) \Delta\chi \quad (9)$$

where χ_0 is an initial value of the contrast, while $\Delta\chi$ is the update of the contrast. $\mathbf{J}_S$ and $\mathbf{H}_S$ are denoted as the Jacobian and Hessian matrices of $S(\chi)$, respectively. $S(\chi)$ can achieve the minimum with its gradient equals to zero, as shown in the following equation:

$$\begin{aligned} \Delta\chi &= -\mathbf{H}_S(\chi)^{-1} \mathbf{J}_S(\chi) = -2\mathbf{H}_S(\chi)^{-1} \mathbf{J}_F(\chi)^T (F(\chi) - E_{obs}^s) \\ &= \mathbf{K}(F(\chi) - E_{obs}^s) \end{aligned} \quad (10)$$

where $\mathbf{J}_F$ is the Jacobian matrix of the forward modeling operator $F(\cdot)$.

In fact, (10) has to be iteratively solved to approach the minimum of $S(\chi)$, because the EM forward process $F(\chi)$ is a nonlinear process and cannot be approximated by the linear operation.

In the conventional method (e.g., Gauss–Newton), $\mathbf{K}$ will be computed in every iteration of a specific case, which lead to extremely huge computation cost. Instead of calculating the matrix $\mathbf{K}$ case by case, the SDM framework only computes the average descent $\mathbf{K}$ for each iteration in the offline training process and directly uses the “learnt” descent $\mathbf{K}$ in the online application process to avoid tremendously huge computation cost. The specific cost function of this offline training process can be described as follows:

$$\min : g(\mathbf{K}_k) = \sum_{n=1}^{N_d} \|\Delta\chi_{k,n} - \mathbf{K}_k \Delta E_{k,n}^s\|_2^2 \quad (11)$$

where $\Delta\chi_{k,n} = \chi_{t,n} - \chi_{k,n}$, $\Delta E_{k,n}^s = E_{k,n}^s - E_{t,n}^s$, and $\chi_{k+1,n} = \chi_{k,n} + \mathbf{K}_k \Delta E_{k,n}^s$. $\chi_{k,n}$ is the optimized contrast at the k th iteration, while $E_{k,n}^s$ is its corresponding optimized computation scattered field. Meanwhile, $\chi_{t,n}$ and $E_{t,n}^s$ are the forward modeling data for training.

In this process, the N_d training models can be rewritten as matrices $\mathbf{X}_t$ and $\mathbf{X}_k$ with N rows, while the corresponding

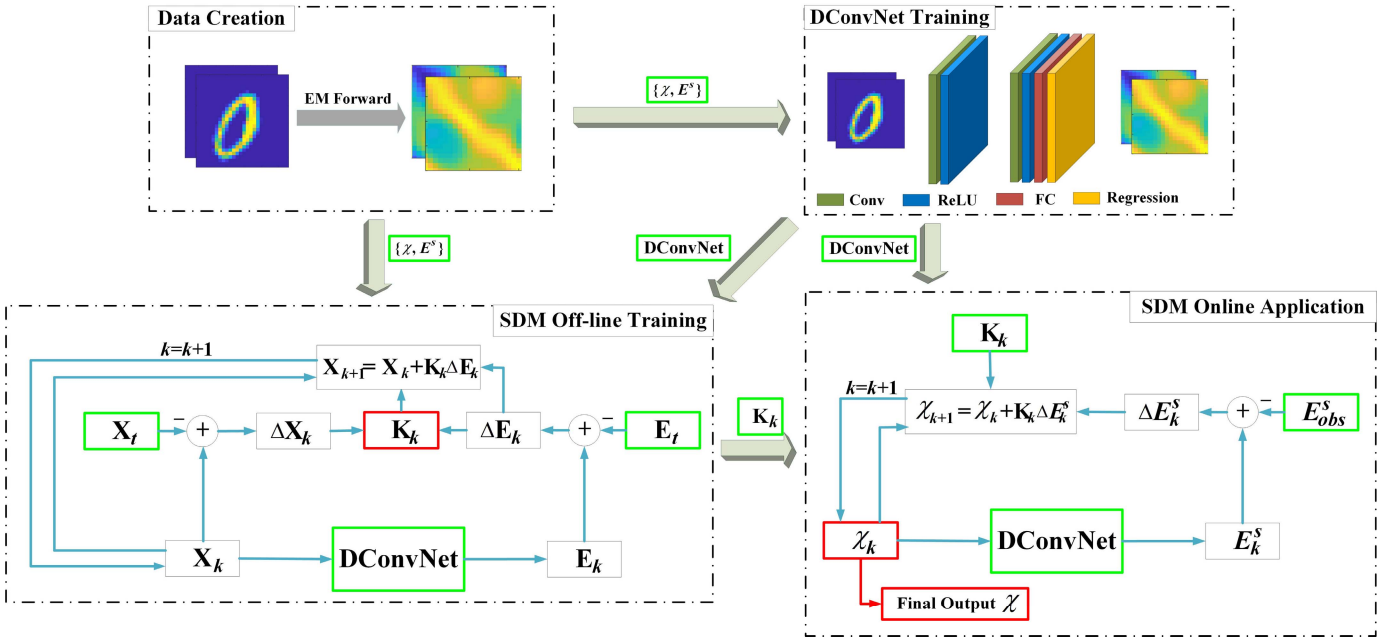


Fig. 3. Workflow of the SDM framework integrated by DConvNet.

forwarding modeling data of these models are denoted as matrices $\mathbf{E}_t$ and $\mathbf{E}_k$. Therefore, the cost function (11) can be rewritten as follows:

$$\mathcal{G}_k = \sum_{n=1}^N \|\Delta \mathbf{X}_k - \Delta \mathbf{E}_k \mathbf{K}_k\|_{\mathcal{F}}^2 \quad (12)$$

where $\Delta \mathbf{X}_k = \mathbf{X}_t - \mathbf{X}_k$ and $\Delta \mathbf{E}_k = \mathbf{E}_k - \mathbf{E}_t$, while $\mathcal{F}(\cdot)$ is the matrix Frobenius norm.

Based on (12), we can finally obtain $\mathbf{K}_k$ as follows:

$$\mathbf{K}_k = (\Delta \mathbf{E}_k^T \Delta \mathbf{E}_k)^{-1} \cdot (\Delta \mathbf{E}_k^T \Delta \mathbf{X}_k). \quad (13)$$

The singular value decomposition (SVD) with a damping factor α is applied to increase the stability of the solving procedure. Thus, $\mathbf{K}_k$ can be finally written as follows:

$$\mathbf{K}_k = \mathbf{V}_k (\mathbf{A}_k^T + \alpha^2 \mathbf{I})^{-1} \mathbf{A}_k^T \mathbf{U}_k^T \Delta \mathbf{X}_k \quad (14)$$

where we have the SVD of $\Delta \mathbf{E}_k$ as $\Delta \mathbf{E}_k = \mathbf{U}_k \mathbf{A}_k \mathbf{V}_k^T$.

Besides, the chosen model misfit (loss) for the offline training between the optimized $\mathbf{X}_k$ and the training true label $\mathbf{X}_t$ can be written as (6). During the offline training process, chunks of matrices $\mathbf{K}_k$ are finally obtained and stored for online application, which are used as descent directions.

In the online application process, the computation process of both the contrast and the scattered field can be written as

$$\begin{cases} \chi_{k+1} = \chi_k + \mathbf{K}_k \Delta \mathbf{E}_k^s \\ \Delta \mathbf{E}_k^s = \mathbf{E}_k^s - \mathbf{E}_{obs}^s \end{cases} \quad (15)$$

where $\mathbf{E}_{obs}^s$ is the received scattered field, χ_k is the updated contrast, and $\mathbf{E}_k^s$ is the corresponding computed scattered field.

However, the biggest challenge of SDM is the large computation cost resulting from the operation $F(\chi)$ in the iteration optimization process. The conventional solving approaches for the operation $F(\chi)$, such as method of moments (MoM),

greatly increase the computation cost of offline training and make online application of SDM difficult in real-time [37], [50]. To overcome this challenge, the proposed DConvNet is integrated into the SDM framework both in offline training and in online application processes. Specifically speaking, the trained DConvNet replaces the operation $F(\chi)$ in the SDM framework. In this way, the computation cost can be greatly reduced, as presented in Section II-D. Another issue we need to emphasize is that the proposed SDM framework integrated by DConvNet can “learn” to correct the error resulting from DConvNet, because the SDM learning process, i.e., the offline training process, is based on $\mathbf{E}^s$ produced by DConvNet and includes its error into the matrix $\mathbf{K}_k$ for the SDM iteration optimization process. Thus, the proposed SDM framework integrated by DConvNet can avoid error accumulation and provide satisfactory result.

D. Workflow for SDM Framework Combined With DConvNet

The workflow of the SDM framework integrated by DConvNet consists of four functional blocks: 1) Data Creation; 2) DConvNet Off-line Training; 3) SDM Off-line Training; and 4) SDM Online Application. The first functional block provides the data for off-line training in the second and third functional blocks, while the trained DConvNet in the second functional block and the trained SDM in the third functional block are combined together to make up the proposed final framework in the fourth functional block, as presented in Fig. 3. The details of each functional block are presented as follows.

1) *Data Creation*: In this functional block, the EM simulation method will be used to create the training data for off-line training of both DConvNet and SDM. While collecting a large number of experimental-based data is difficult for researchers, preparing simulation-based data is more flexible and efficient, because they can be constructed based on various synthetic

samples and can be accurately computed by some numerical methods in a relatively accepted time period. We use MoM to solve (1) and (2) and compute the scattered fields illuminated by an incident EM wave. As presented in the setups in Fig. 1, N_i groups of scattered fields are obtained by M receivers with N_i incident waves, respectively, illuminating certain scatterers. For the conventional methods, N_i groups of scattered fields will be used to reconstruct the contrast (permittivity) of the scatterer by the optimization process in (3). These training data will be shared by off-line training of both DConvNet and SDM. Thus, the computation cost for creating training data in the newly proposed framework will not be increased, compared with the conventional SDM [37], [50].

To demonstrate the capability and generalization ability of the proposed method, only one group of synthetic dataset, i.e., MNIST dataset [32], [33], [34], [35], is chosen as the scatterer to create the training and testing datasets. Using the MNIST dataset, the digit-shaped scatterers are given the inhomogeneous relative permittivity. A group of samples in the MNIST dataset are randomly chosen to create the training data, i.e., χ and E^s , for both DConvNet and SDM training, while some new samples are chosen to create testing data for both DConvNet and SDM.

2) *DConvNet Off-Line Training*: In this functional block, the proposed DConvNet directly makes use of the data, i.e., χ and E^s , created by MNIST in the previous functional block as the input and output to do training and testing, as shown in Fig. 3. The specific training process follows the described details in Section II-B, while the generated χ and E^s are its input and the corresponding output, respectively. One example of its testing performance is presented in Section III-B. The trained DConvNet can predict E^s (i.e., all the scattered fields in the measurement setting) of a scatterer at one time and can greatly reduce the computation cost of the operation $F(\chi)$ with high accuracy for the EM forward process, as the performance presented in Section III-B. Ultimately, the trained DConvNet will greatly improve the efficiency for both the SDM offline training and its online application processes, as presented in Fig. 3.

Moreover, the newly proposed DConvNet model is implemented into MATLAB 2019a with the DL Toolbox [46] for offline training. Adaptive Moment Estimation (Adam) optimizer is used for optimizing the mentioned loss function. Compared with other optimizers, including stochastic gradient descent (SGD), Adam optimizer generally navigates through the loss surface more favorably [47].

3) *SDM Off-Line Training*: The detailed procedures of SDM offline training follows the mechanism described in Section II-C, while the created training data (i.e., χ and E^s) and the trained DConvNet will be used during this process. The aim of SDM offline training is to compute the average descent $\mathbf{K}_k$ in (11) for each iteration in the optimization process. This average descent $\mathbf{K}_k$ will be used to predict the contrast (permittivity) of the unknown scatterer. Unlike the conventional methods (e.g., Gauss-Newton), both the offline training and online application of SDM never require to compute Jacobian or Hessian matrices in (10) during the optimization process. However, one challenge for the conventional SDM offline training is the large computation cost resulting from the operation $F(\chi)$ in the iteration optimization

TABLE II
PERFORMANCE COMPARISON OF SDM AND SDM WITH DCONVNET

	DConvNet Training	SDM Offline Training	Reconstruction on MNIST	Reconstruction on Cylinder
SDM with DconvNet	648.5s	1646.8s	0.4606s	0.4581s
SDM		86351.3s	24.89s	26.10s
Gauss-Newton			162.45s	161.17s

process. The conventional solving approaches of $F(\chi)$, e.g., MoM, greatly increase the computation cost of offline training. Thanks to the combination of the trained DConvNet, SDM offline training in the proposed framework can be done with less computation cost and CPU time. Specifically speaking, the trained DConvNet replaces the operation $F(\chi)$ in the SDM framework, which greatly reduces its computation cost. Besides, the error from the trained DConvNet can also be “learnt” into the SDM framework, so that the error from the trained DConvNet can also be corrected during the online application process, as presented in Sections III-C and III-D. The comparison between the conventional SDM training and SDM offline training in the proposed framework is presented in Table II. Ultimately, based on the trained DConvNet and the created data (i.e., χ and E^s) in the first functional block, the average descent $\mathbf{K}_k$ is “learned” and further used for online application of the proposed ML framework.

4) *SDM Online Application*: Based on the trained DConvNet and the continuous group of the average descent direction $\mathbf{K}_k$ computed from the SDM offline training process, the newly proposed ML framework can solve EMIS problem for a new scatterer case with high accuracy in real-time. The specific process of online application is that the initial contrast χ_0 of the scatterers is input to the proposed DConvNet, while its output field E^s is used by SDM to do optimization computation. Then, the obtained $\Delta\chi$ is used to update the contrast. By repeating this process iteratively, the final optimized contrast χ is obtained. The specific performance of this framework is presented in Section III. Because of the embedding of our DConvNet to replace the conventional EM forward computation, the computation cost of SDM for online application has been greatly reduced and can even fulfill the requirement of real-time application, as shown in Table II. The specific performance of the proposed ML framework for solving EMIS problem is demonstrated in Section III-B.

III. NUMERICAL EXAMPLES

In this section, the specific off-line training and testing of the proposed approach are discussed. We emphasize that the training of both DConvNet and SDM is based on only one group of synthetic dataset, i.e., MNIST dataset [32], [33], [34], [35]. Meanwhile, we use several numerical examples to demonstrate the precision and effectiveness of the newly proposed DL approach.

A. Numerical Setup

In D_{obj} shown in Fig. 1, the size of each sample in the MNIST dataset is set to about $\lambda \times \lambda$ (the wavelength $\lambda = 1$ m in free space) with the relative permittivity ϵ_r , while each sample is gridded into 28×28 uniform pieces (i.e., $N = 28$).

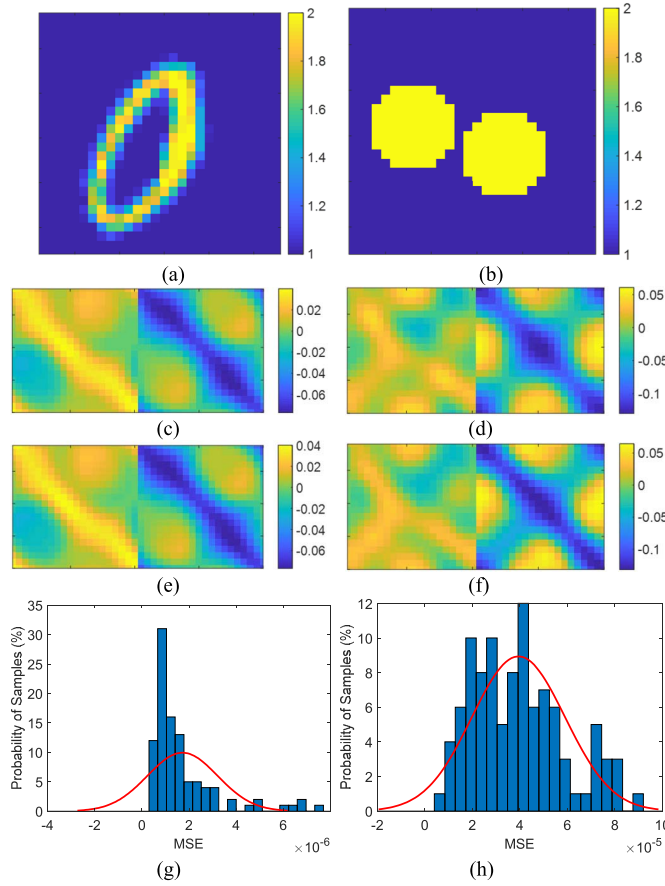


Fig. 4. Typical examples in two synthetic datasets. (a) MNIST. (b) Cylinder. (c) Ground-truth field “image” from the sample in (a) (Left: real part; Right: imaginary part). (d) Ground-truth field “image” from the sample in (b) (Left: real part; Right: imaginary part). (e) Output of DConvNet for the sample in (a) (Left: real part; Right: imaginary part). (f) Output of DConvNet for the sample in (b) (Left: real part; Right: imaginary part). The error analysis for both (g) MNIST and (h) Cylinder.

While 20 receivers ($M = 20$) are uniformly set around D_{obj} with the distance 20λ , the TM_z incident plane waves come from 20 different directions to impinge on the object domain D_{obj} with the incident angles uniformly distributed within $(0^\circ, 360^\circ)$ (i.e., $N_i = 20$). In this section, the full-wave EM simulations [51] are used for all the samples to create both the training and testing datasets.

Using the MNIST dataset, the digit-shaped scatterers are given the inhomogeneous relative permittivity ε_r ranging from 1 to 2, as presented in Fig. 4(a). In all, 5000 samples in the MNIST dataset are randomly chosen to create the training dataset, i.e., χ and E^s , for both DConvNet and SDM training. Offline training includes two steps. First, the training of DConvNet follows the described details in Section II-B, while the generated χ and E^s are its input and the corresponding output, respectively. Its testing performance is presented in Section III-B. The trained DConvNet can greatly reduce the computation cost of the operation $F(\chi)$ with high accuracy for the EM forward process, presented in Section III-B. In the second step, the trained DConvNet is integrated into the SDM framework, while the average descent directions between the initial prediction and the true label of SDM iterative schemes are learned using the same dataset in the first step. The detailed

procedures follow the described computation mechanism in Sections II-C and II-D. Because of the embedding of our DConvNet to replace the conventional EM forward computation, the computation cost of SDM for both training and application has been greatly reduced. Besides, the error from the trained DConvNet can also be “learnt” into the SDM framework, so that this error can be corrected during online application and testing, as presented in Sections III-C and III-D. Both the two-step offline training and their application are implemented into MATLAB 2019a with the DL Toolbox [46]. To evaluate the result of retrieved “images” in Sections III-C and III-D, two quantitative indicators, i.e., normalized mean square error (NMSE) and structural similarity index (SSIM), are used, as what has been done in [32], [33], [34], and [35].

B. Performance of DConvNet for EM Forward Process

In this section, the performance of the trained DConvNet for the EM forward process is evaluated by two kinds of database. First, the new unknown object in the MNIST database is used as the tested scatterer shown in Fig. 4(a), while the corresponding field “image” with the real and imaginary parts, i.e., E^s , is presented in Fig. 4(c) as the ground truth. As the comparison, the field “image” computed from the trained DConvNet in Fig. 4(e) is much approximated to its ground truth, which indicates the precision and effectiveness of the newly proposed DConvNet for the EM forward process.

In the second example, a new synthetic dataset, called “Cylinder,” is proposed, where two homogeneous cylinders are assumed to be randomly located in D_{obj} as scatterers. The diameters of two cylinders in “Cylinder” are both set as $\lambda/3$, while their relative permittivity ε_r is set as 2. We here emphasize that the overlapping between two cylinders is allowed. One testing sample is presented in Fig. 4(b), while its corresponding field “image” in Fig. 4(d) is used as the ground truth. Although the proposed ConvNet is based on the MNIST dataset, its test on the totally new “Cylinder” dataset illustrates its strong capacity of modeling the EM forward process. As shown in Fig. 4(f), there is nearly no difference between itself and the ground truth in Fig. 4(d).

Fig. 4(g) and (h) presents the analysis of loss between the ground-truth field “image” E^s and the corresponding predicted field “image” from the trained DConvNet based on 5000 samples in two mentioned synthetic datasets. The average of mse in (14) for two datasets is around 10^{-6} and 10^{-5} , respectively, which illustrates the accuracy of DConvNet. Besides, the computation time of DConvNet is only about 0.034 s, while the conventional method (i.e., MoM) will take about 2.262 s to complete the EM forward process. Thus, due to the combination of DConvNet, the computation cost of the proposed SDM framework can be greatly reduced, compared with the conventional methods for EMIS problems and the conventional SDM [37], [50].

Then, the trained DConvNet is integrated into the SDM framework to obtain the average descent directions to solve the EMIS problem for further online application.

C. Reconstruction of Number-Shaped Scatterers

In this section, the trained SDM framework with DConvNet is tested on 5000 new samples in the MNIST dataset.

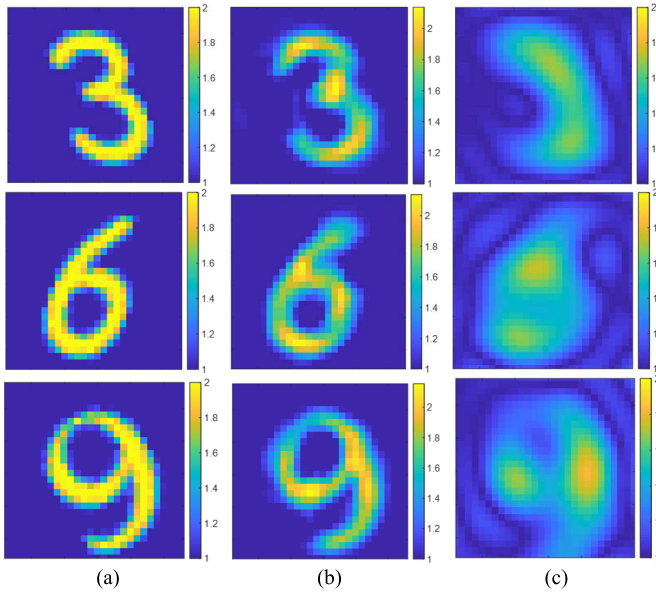


Fig. 5. Comparisons of reconstructed relative permittivities of digit-shaped scatterers in MNIST. (a) Ground truth. (b) Reconstructed relative permittivities from the proposed SDM with DConvNet. (c) Reconstructed relative permittivities from the Gauss–Newton method.

To indicate the power of the proposed approach, 5% Gaussian noise is added into the scattered field, i.e., E^s , produced from 5000 testing samples. NMSE and SSIM are still used to evaluate the result of retrieved “images,” as [32], [33], [34], and [35].

Fig. 5 demonstrates the comparison between the ground truth, the reconstructed results of the Gauss–Newton method, and the reconstructed results of the proposed SDM with DConvNet. Evidently, the final reconstruction outputs of the proposed method have good agreement with the ground truth, though the new test shape and inhomogeneous relative permittivity. However, the conventional method (Gauss–Newton method) can hardly provide satisfactory result. Meanwhile, the proposed ML-based method takes much less time to realize reconstruction than the Gauss–Newton method, as shown in Table II.

In Fig. 6(a) and (b), the statistical analyses for the testing results after ten iterations are presented: the NMSE average of the reconstructed results obtained by DConvNet is around 0.01, and the SSIM average obtained from the proposed approach can overcome 0.8. Therefore, the newly proposed approach can resolve the EMIS problem with excellent performances. Besides, the average change of both NMSE and SSIM is shown in Fig. 6(c) and (d), where the average of NMSE and SSIM is gradually improved to around 0.01 and 0.8, respectively. The specific comparison between the conventional SDM and the proposed SDM with DConvNet can also be found in Table II.

Meanwhile, we also analyze the noise sensitivity for our proposed method. We set 5%, 10%, 20%, 30%, and 40% Gaussian noise to the input of the our proposed framework. As shown in Fig. 7, our proposed approach can demonstrate excellent noise tolerance. When the level of Gaussian noise achieves 40%, our proposed approach can still provide meaningful reconstruction result, while the reconstruction result can

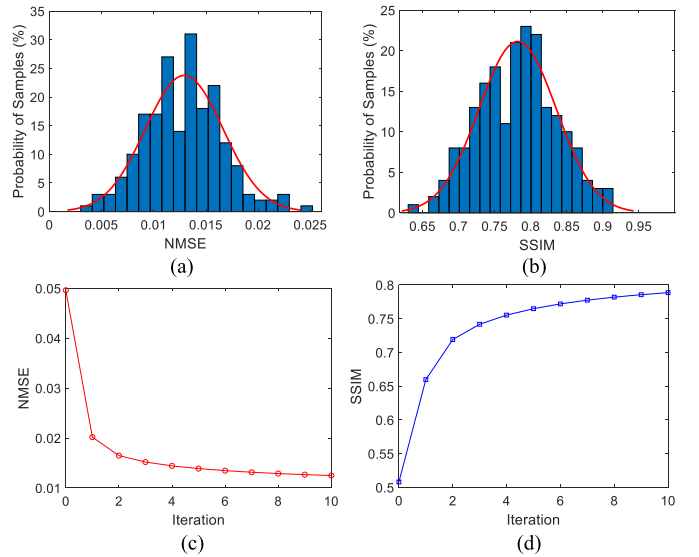


Fig. 6. Performance of the proposed SDM with DConvNet on the MNIST dataset. Statistical histograms of the image quality for the reconstructed permittivities and fitting of its normal density function. (a) NMSE results from the proposed SDM with DConvNet. (b) SSIM results from the proposed SDM with DConvNet. The improvement of image quality for the reconstructed permittivities with iteration. (c) Average NMSE with iteration. (d) Average SSIM with iteration.

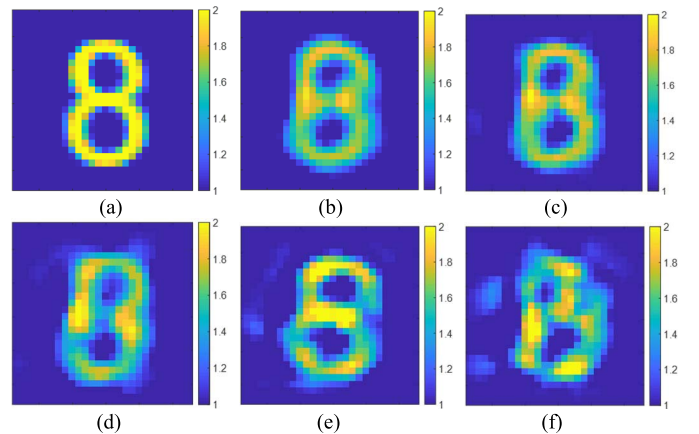


Fig. 7. Comparisons of reconstructed relative permittivities of digit-shaped scatterers in MNIST with different noise levels. (a) Ground truth. (b)–(f) Gauss noise: (b) 5%, (c) 10%, (d) 20%, (e) 30%, and (f) 40%.

be much approximated to the ground truth with the noise level below 30%.

D. Reconstruction of “Cylinder” Scatterers

As Section III-C, the trained SDM framework with DConvNet is tested on 5000 samples in the “Cylinder” dataset. While 5% Gaussian noise is added into the scattered field, i.e., E^s , produced from 5000 testing samples, NMSE and SSIM are applied to evaluate the result of reconstructed images. Here, we stress that training of both DConvNet and SDM is only based on samples in MNIST rather than “Cylinder.”

Fig. 8 illustrates the comparison of one example between the ground truth, the reconstructed results of the Gauss–Newton method, and the reconstructed results of the proposed SDM with DConvNet. Considering that the training of both

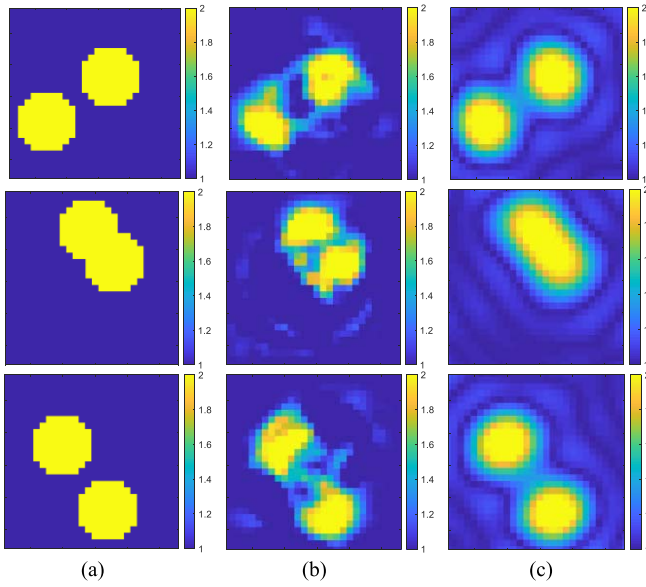


Fig. 8. Comparisons of reconstructed relative permittivities of cylinder-shaped scatterers in Cylinder. (a) Ground truth. (b) Reconstructed relative permittivities from the proposed SDM with DConvNet. (c) Reconstructed relative permittivities from the Gauss–Newton method.

DConvNet and SDM is based only on the MNIST dataset, the final reconstruction outputs of the proposed method provide satisfactory results for solving EMIS problem. Besides, while our proposed SDM with DConvNet can offer similar results with the Gauss–Newton method, our proposed approach takes much less time to realize reconstruction than the Gauss–Newton method which has to waste a huge amount of time in solving the EM forward process. The comparison between our proposed approach and the Gauss–Newton method is presented in Table II. Meanwhile, the result also indicates the strong generalization ability of the proposed method. Although the training of the proposed is based on one dataset, the trained SDM with DConvNet can solve the EMIS problems for another totally different dataset.

Fig. 9(a) and (b) illustrates the statistical analyses for the testing results at the tenth iteration: the NMSE average of the reconstructed results obtained by DConvNet is around 0.03, while the SSIM average obtained from the proposed approach can overcome 0.6, which demonstrates that the proposed approach can successfully resolve the EMIS problem. Furthermore, the tendency of the average for both NMSE and SSIM is shown in Fig. 9(c) and (d), where the average of NMSE and SSIM is finally improved to around 0.035 and 0.6, respectively. The specific comparison between the conventional SDM and the proposed SDM with DConvNet can also be found in Table II.

E. Reconstruction of Experimental Data

To further demonstrate the capability of the proposed framework, we challenge the proposed ML framework by the popular experimental data measured at Institute Fresnel [34], [37], where the transmitting and receiving antennas are both wideband-ridged horn antennas, and the source–object center and object–receiver distances are 1.67 m. Accordingly, eight transmitters and 241 receivers are used. This experiment-based

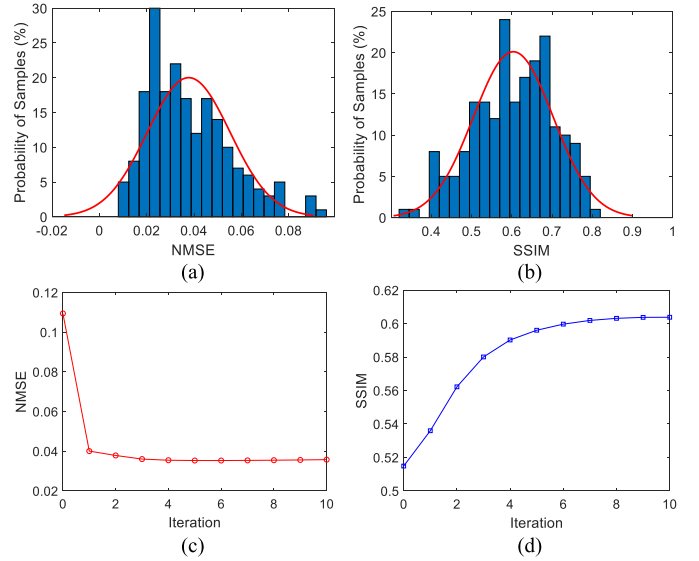


Fig. 9. Performance of the proposed SDM with DConvNet on the Cylinder dataset. Statistical histograms of the image quality for the reconstructed permittivities and fitting of its normal density function. (a) NMSE results from the proposed SDM with DConvNet. (b) SSIM results from the proposed SDM with DConvNet. The improvement of image quality for the reconstructed permittivities with iteration. (c) Average NMSE with iteration. (d) Average SSIM with iteration.

scenario (i.e., the “FoamDieInt” and “FoamDieExt”) consists of two cylinders: the big cylinder has a diameter of 0.08 m with a relative permittivity of $1.45 (\pm 0.15)$, while the small cylinder has a diameter of 0.031 m with a relative permittivity of $3 (\pm 0.3)$.

To fit the experimental data from Fresnel and illustrate the generalization ability of the proposed method, we change the dimension of the input and output of the proposed DConvNet for the EM forward process. Thus, E^s is the input of DConvNet with $M = 241$ and $N_i = 8$, while χ has the dimension $32 \times 32 \times 2$ with $N = 32$. Then, we create the training data and perform offline training for both DConvNet and SDM with the same process described in Section II-D and further test the trained ML framework for the experimental data. We emphasize that only MNIST dataset is still used to create the simulation-based training data for offline training of both DConvNet and SDM. Again, 5000 samples in the MNIST dataset are randomly chosen to create the training dataset, i.e., χ and E^s , for both DConvNet and SDM training, while the digit-shaped scatterers are given the inhomogeneous relative permittivity ε_r ranging from 1 to 3.

The performance of the trained ML framework is presented in Fig. 10. While the unknown digit-shaped scatterers in the MNIST dataset are successfully reconstructed again, the proposed approach also provides the acceptable reconstruction result for the experimental data.

To further explore the dependence of our proposed ML approach on the training dataset, we change the training dataset from MNIST to a new synthetic dataset similar to the experimental scenario, which consists of two cylinders: one cylinder has a random diameter ranging from 0.06 to 0.1 m with the random relative permittivity ranging from 1.2 to 1.8, while another cylinder a random diameter ranging from 0.02 to 0.04 m with the random relative permittivity ranging

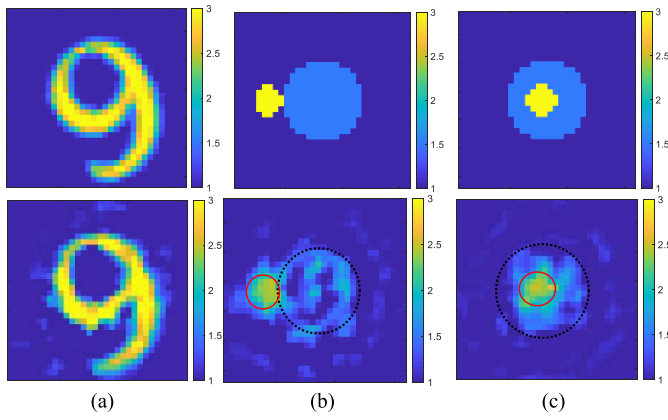


Fig. 10. Comparisons between the ground truth and reconstructed relative permittivities from the newly trained framework. (a) Number-shaped scatterer. (b) FoamDielExt. (c) FoamDielInt.

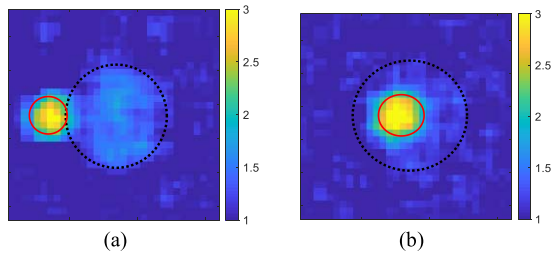


Fig. 11. Performance of the proposed ML framework on the new synthetic dataset. (a) FoamDielExt. (b) FoamDielInt.

from 2.5 to 3.5. The two cylinders are randomly located in the objective domain as scatterers, while the overlapping between them is allowed. We choose again 5000 groups of training data according to this new kind of synthetic dataset to train our proposed ML framework and test the trained framework with the experimental data. As the observation in [37] and [50], Fig. 11 presents that the reconstruction result based on the new synthetic dataset is more approximated to the ground truth, compared with the trained framework on the MNIST dataset. Generally, if the training dataset is similar to the testing dataset, the trained DL model will make use of more prior information in training to do better prediction. Thus, this characteristic about training dependence can act as an important issue to design the training dataset to further improve the performance of our proposed framework.

F. Discussion

From Sections III-A–III-E, we can find that the accuracy of reconstruction of our proposed ML framework can be improved with more “accurate” prior information integrated during training. For our proposed approach, the prior information are ultimately integrated into two parts: the trained DConvNet for the EM forward process and the computed average descent $\mathbf{K}_k$ of SDM for the EMIS process. Generally speaking, if the training dataset has high similarity to the testing dataset (i.e., the training and testing have similar solution space), the learning-based methods [32], [33], [34], [35], [37], [50] for EMIS can better make use of “prior information” resulting from the similarity and further solve EMIS problems with high accuracy and even high resolution for EM imaging. However, if the training dataset shares less similarity with the testing dataset, the obtained “prior information” during training for the learning-based methods will hardly help improve the accuracy for solving EMIS

problem. For online application, our proposed ML framework makes use of the trained DConvNet and the trained SDM to alternatively solve the EM forward and EM inverse processes. In this process, the feedback from the trained model and that from the observed data are alternatively updated by the trained parameters of our proposed ML framework, which also help decrease overfitting.

Based on the benchmarks in this article, our proposed ML framework can solve the EMIS problem with high computation efficiency for any cases, and it can even meet the requirement of online application, as shown in Section III-C. As for the accuracy of reconstruction, our proposed ML framework can solve the EMIS problem with high accuracy and even extremely high resolution for EM imaging, when sufficient prior information can be made use of, as shown in Section III-C. Moreover, if sufficient prior information cannot be provided, our proposed ML framework can demonstrate the high generalization ability based on the training dataset with the relative weak prior information, but it will have to sacrifice the privilege of solving EMIS problem with extremely high accuracy and even high resolution, which has been illustrated in Sections III-D and III-E. Furthermore, with extremely limited prior information, the accuracy of our proposed ML framework will be degraded. But it can be perceived by monitoring the data misfit during the online application process. When the data misfit fails to reduce, the online application of the proposed ML framework should be paused. If it happens, more “accurate” prior information has to be added to improve the performance of the proposed ML framework or the conventional methods for EMIS problems could also be considered for this special case.

IV. CONCLUSION

In this article, a novel DL framework is proposed to solve EMIS problems. A complex-valued DConvNet is integrated into the SDM to realize both the offline training and the online application for EMIS problems. The offline training includes two steps. While DConvNet is trained to realize the EM forward process, SDM with this trained DConvNet is trained to compute the average descent directions between the initial prediction and the true label of the SDM iterative schemes based on the same dataset. Therefore, the prior information for EMIS process is “learnt” into this framework to improve future application. In the online application step, the contrasts (permittivities) of scatterers are reconstructed by the SDM iteration process using the learned descent directions, while the EM forward process is repeatedly realized by the trained complex-valued DConvNet. Consequently, this framework can successfully integrate the prior information into the EMIS solving process with satisfactory accuracy and efficiency. Besides, the newly proposed framework has low computational complexity and can realize imaging in real-time. Numerical examples demonstrate the performance of the novel proposed framework.

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